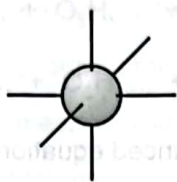
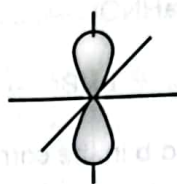


- 6 The diagrams show the shape of two orbitals, with principal quantum number 3, in an isolated atom.



X



Y

Which orbital has the lower energy and how many of this type of orbital have the principal quantum number 3 in a single isolated atom?

	orbital with lower energy	number of orbitals of that type with principal quantum number 3
A	X	1
B	X	3
C	Y	1
D	Y	3